
Algorithm 1 Event-driven simulation subroutine

Require: Γ : ordered tasks, runtimes, object sizes/locations, and movement triggers

$\mathcal{H}$: memory capacities, directional bandwidths, and deterministic event semantics

Ensure: Validity $v \in \{\text{false}, \text{true}\}$ and makespan $m \in \mathbb{R}_{\geq 0} \cup \{\infty\}$

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1: if  $\neg \text{StaticValid}(\Gamma)$  then
2:   return (false,  $\infty$ ) ▷ Reject malformed plans
3: end if
4:  $\Sigma \leftarrow \text{Initialize}(\Gamma, \mathcal{H})$ 
5: for all tasks  $\tau_i$  in declared order do
6:   while  $\neg \text{Ready}(\Sigma, \tau_i)$  do
7:     if  $\neg \text{AdvanceNextEvent}(\Sigma, \mathcal{H})$  then
8:       return (false,  $\infty$ )
9:     end if
10:  end while
11:   $\Sigma \leftarrow \text{ExecuteTask}(\Sigma, \tau_i, \mathcal{H})$  ▷ Reserve outputs; overlap transfers
12: end for
13: while  $\text{TransfersPending}(\Sigma)$  do
14:   if  $\neg \text{AdvanceNextEvent}(\Sigma, \mathcal{H})$  then
15:     return (false,  $\infty$ )
16:   end if
17: end while
18: if  $\neg \text{FinalStateValid}(\Sigma)$  then
19:   return (false,  $\infty$ )
20: end if
21: return (true,  $\text{Makespan}(\Sigma)$ ) ▷ All required final copies are current
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